NETWORKING REFERENCE & TOOLS

Networking Reference & Tools — Intermediate Networking

Version 1.3 • Last reviewed: 12 October 2026

FIELD GUIDE / STUDY REFERENCE

PRACTICAL REFERENCE · VERSION 1.3

INTERMEDIATE NETWORKING EDITION • Subnetting • VLANs and routing • Network services • Structured troubleshooting

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DESIGN GOAL

A compact companion for networking students: fast to scan, easy to revisit, and focused on applying knowledge. It complements formal courses rather than replacing them.

Choose the right depth

Basic: identify the likely fault layer and gather first-response evidence. **Intermediate:** isolate VLAN, routing, DHCP/DNS and packet-level faults. **Advanced:** reason across routing, policy, IPv6, operations and change control.

Jump to the troubleshooting flow or the edition-specific workbench.

How to use this reference

This edition assumes you understand basic IP addressing, Ethernet, DNS, DHCP, and the OSI and TCP/IP models. It focuses on diagnosing and working with connected networks rather than learning every term from scratch.

Use the playbooks as an evidence-led sequence. Confirm the scope of an issue, collect command output, test one hypothesis at a time, and document changes.

Advanced routing protocols, deeper switching, network automation, and full CCNA exam preparation are reserved for the separate Advanced Networking & CCNA Preparation edition.

Document revision: 1.3 • Last reviewed: 12 October 2026. Verify platform-specific syntax and current vendor guidance before making changes.

This toolkit was developed with assistance from AI tools for drafting, organisation and refinement. The content was reviewed by the author against their own networking knowledge and learning, with the aim of improving accuracy, clarity and practical usefulness. The resource is subject to ongoing technical review, fact-checking against authoritative documentation, and refinement based on external feedback. Readers should verify critical procedures against current vendor documentation and test configurations in an appropriate lab or authorised environment.

Suggested order: scan the relevant cheat sheet → run the simplest useful check → compare the result with the playbook → record the finding.

Daily-use quick reference

Start with scope and evidence. The quickest useful test is the one that separates likely causes without changing the system.

Repeatable workflow: Define the symptom → identify who/what is affected → inspect link and address configuration → test gateway → test a known IP → test DNS/name resolution → inspect the route and service → change one thing only with approval → verify and record.

Choose the right depth

Symptom	First checks	Next useful distinction
One device offline	Adapter/link, Wi-Fi association, IP/prefix, gateway, DNS	Compare with a working device on the same network.

Symptom	First checks	Next useful distinction
169.254.x.x IPv4 on a DHCP client	DHCP lease, VLAN/SSID, DHCP server/relay, scope exhaustion	Check whether other clients on the same segment obtain leases.
IP works; name fails	Configured DNS servers, resolver response, suffix/search policy	Compare a known-good name and a direct IP test; do not treat one ping result as conclusive.
Local subnet works; remote network fails	Default route, next hop, routing table, ACL/firewall, return path	Use traceroute/tracert and device-side route checks where authorised.
Wi-Fi connected but slow or intermittent	Scope of impact, signal, band, channel use, roaming, driver, AP/controller events	Compare near the AP, another client, and wired service where possible.
One service/port fails	Correct hostname/IP, port, protocol, service state, listening socket	Separate name resolution, route, firewall, TCP reachability, TLS and application health.
Multiple users/sites affected	Time of onset, shared dependency, recent changes, monitoring/alerts	Escalate early with evidence; avoid isolated endpoint resets that destroy clues.

Escalate promptly for suspected security incidents, broad outages, repeated production failures, unknown/unauthorised configuration changes, risk of data loss, or when the next step exceeds your access, authority, change window or rollback capability.

1. Quick-reference cheat sheets

1.1 Network devices and their jobs

1.1 Network devices and their jobs

Component	Primary function	Useful reminder
NIC / network adapter	Connects a device to Ethernet or Wi-Fi.	Check interface state, driver, link, and assigned configuration.
Repeater / hub	Repeats signals; a traditional hub sends incoming bits to all other ports.	Mostly legacy; hubs do not learn MAC addresses.
Layer 2 switch	Forwards Ethernet frames based on learned MAC addresses and VLAN membership.	Check port state, VLAN, MAC table, and trunking.
Router	Forwards IP packets between networks using a routing table.	Check route, next hop, interface state, and policy.
Layer 3 switch	Combines switching with routing between VLANs/subnets.	SVIs and routing must be configured and operational.
Wireless access point	Bridges wireless clients to a wired LAN.	SSID, authentication, radio conditions, VLAN mapping, and uplink matter.
Firewall	Allows or denies traffic according to security policy.	Check direction, source/destination, protocol, port, state, and rule order.

Component	Primary function	Useful reminder
Modem / ONT	Terminates a provider access technology.	Link synchronisation does not guarantee working internet service.
DNS resolver	Finds DNS records for client queries, often using cache and upstream servers.	Separate client-to-resolver reachability from resolver-to-authoritative resolution.
DHCP server	Leases addresses and provides network settings.	Scope, exclusions, lease, relay, and VLAN path matter.

1.2 OSI model at a glance

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Layer	Name	Examples / troubleshooting focus
7	Application	HTTP(S), DNS, DHCP; application settings and service responses.
6	Presentation	Data representation, encoding, encryption formats; often folded into application implementations.
5	Session	Dialog/session management; conceptual separation varies in real stacks.
4	Transport	TCP, UDP; ports, connection setup, retransmission, listening service.

Layer	Name	Examples / troubleshooting focus
3	Network	IPv4/IPv6, ICMP, routing; IP addressing, masks, routes.
2	Data Link	Ethernet, Wi-Fi MAC, VLANs, ARP for IPv4 neighbour mapping; frames and switching.
1	Physical	Copper, fibre, radio, connectors, signal, link state.

1.2 OSI model at a glance (part 2)

Use OSI as a diagnostic model Start with the lowest plausible failure, but do not force every problem into exactly one layer. Real faults can cross layers.

1.3 TCP/IP model

1.3 TCP/IP model

TCP/IP layer	Typical functions	OSI relationship (approximate)
Application	Application protocols and data formats	OSI 5–7
Transport	End-to-end process communication; TCP/UDP	OSI 4
Internet	IP addressing and packet routing	OSI 3

TCP/IP layer	Typical functions	OSI relationship (approximate)
Link / Network access	Local delivery over Ethernet/Wi-Fi and physical media	OSI 1–2

1.4 Common protocols and systems

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Protocol / system	What it does	Related ideas
IPv4 / IPv6	Logical addressing and packet delivery across networks.	Subnets, routing, gateways, ICMP.
Ethernet	Wired LAN framing and link operation.	MAC addresses, switches, VLAN tags.
ARP (IPv4)	Finds a local-link MAC address for an IPv4 neighbour.	Broadcast domain, ARP cache, gateway MAC.
NDP (IPv6)	IPv6 neighbour discovery and related local-link functions.	ICMPv6, router discovery, address resolution.
ICMP	Error reporting and diagnostic messaging for IP.	ping, traceroute/tracert; may be filtered.
TCP	Reliable, ordered byte stream.	Ports, SYN/SYN-ACK/ACK, retransmissions.
UDP	Datagram transport without built-in delivery guarantee.	DNS, DHCP, NTP, real-time applications.
DNS	Resolves names and stores other domain data.	A, AAAA, CNAME, MX, PTR, TXT records.

Protocol / system	What it does	Related ideas
DHCPv4	Leases IPv4 settings dynamically.	Discover, Offer, Request, Acknowledge (DORA).
HTTP / HTTPS	Web application protocol; HTTPS uses TLS protection.	TCP 80/443 commonly; HTTP/3 uses QUIC over UDP.
SSH	Secure remote shell and tunnelling.	TCP 22 commonly; key-based authentication.
NTP	Synchronises clocks.	Timestamps, certificates, logs, authentication.
SNMP	Monitoring and management of network devices.	Managers, agents, MIBs; secure versions/configuration preferred.
Syslog	Transports or records system log messages.	Severity, central logging, incident timelines.
TLS	Protects data in transit and authenticates endpoints when correctly configured.	Certificates, trust chain, hostname, expiry.
NAT/PAT	Translates IP addresses and often transport ports.	Private IPv4, state tracking, inbound rules.
802.1Q	Adds VLAN tags to Ethernet frames on trunks.	Allowed VLAN list and native VLAN behaviour.
STP/RSTP	Prevents Layer 2 loops by controlling redundant paths.	Root bridge, blocked/discarding ports, topology changes.
OSPF	Link-state interior routing protocol.	Neighbours, areas, LSDB, routes, metrics.

Protocol / system	What it does	Related ideas
BGP	Exchanges routing information between autonomous systems and in some internal designs.	Peering, prefixes, policy, AS path.

1.5 Common ports

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Port / protocol	Service	Notes
20/21 TCP	FTP data/control	Legacy and unencrypted by default; mode affects data connections.
22 TCP	SSH	Secure remote access.
23 TCP	Telnet	Unencrypted; avoid on untrusted networks.
25 TCP	SMTP relay	Mail transfer; client submission often uses 587 or 465 depending on setup.
53 TCP/UDP	DNS	Both transports are used in different cases.
67/68 UDP	DHCPv4 server/client	Address leasing.
69 UDP	TFTP	Simple file transfer; limited security.
80 TCP	HTTP	Web traffic without TLS.
110 TCP	POP3	Email retrieval; secure configurations differ.
123 UDP	NTP	Time synchronisation.

Port / protocol	Service	Notes
143 TCP	IMAP	Email access; secure options available.
161/162 UDP	SNMP query/traps	Access control and version matter.
389 TCP/UDP	LDAP	Directory access; secure deployments use appropriate protections.
443 TCP; UDP 443 for QUIC	HTTPS / HTTP/3	HTTPS commonly uses TCP; HTTP/3 uses QUIC over UDP.
445 TCP	SMB	Windows file/printer sharing; restrict exposure.
514 UDP commonly	Syslog	Transport/security depends on implementation.
587 TCP	SMTP submission	Client-to-server mail submission commonly uses STARTTLS.
636 TCP	LDAPS	LDAP over TLS.
3389 TCP/UDP	RDP	Remote Desktop; tightly restrict access.
3306 TCP	MySQL	Database service; do not expose unnecessarily.
5432 TCP	PostgreSQL	Database service; restrict and authenticate.

1.5 Common ports (part 2)

Port reminder A port number identifies a transport-layer endpoint, not a physical switch port. Service defaults can be changed; confirm the actual configuration.

1.6 IPv4 subnetting pocket reference

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CIDR	Subnet mask	Total addresses	Conventional usable hosts
/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/29	255.255.255.248	8	6
/30	255.255.255.252	4	2
/31	255.255.255.254	2	2 for point-to-point use under RFC 3021
/32	255.255.255.255	1	Not applicable — single-address host route

- Block size in the interesting octet = 256 − the mask value in that octet.
- Subnet boundaries occur at multiples of the block size.
- For conventional IPv4 subnets, identify network, first host, last host, and broadcast.
- /31 point-to-point links and /32 host routes are special cases.
- Cross-check non-trivial calculations with a second method or a trusted calculator.

1.6 IPv4 subnetting pocket reference (part 2)

Worked example	Result
Address	192.168.10.77/26
Mask / block size	255.255.255.192 / block size 64 in the final octet
Boundaries	0, 64, 128, 192
Network	192.168.10.64/26
Host range	192.168.10.65–192.168.10.126
Broadcast	192.168.10.127
Conventional usable hosts	62

1.7 Private IPv4 ranges and special addresses

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Range	Purpose / caution
10.0.0.0/8	Private IPv4 addressing.
172.16.0.0/12	Private IPv4 addressing; 172.16–172.31 in the second octet.
192.168.0.0/16	Private IPv4 addressing.
169.254.0.0/16	IPv4 link-local; often appears on Windows when DHCPv4 fails.
127.0.0.0/8	IPv4 loopback; 127.0.0.1 is the common local test address.
0.0.0.0	Context-dependent; can mean unspecified address or default route notation.

Range	Purpose / caution
255.255.255.255	Limited IPv4 broadcast address.

1.8 VLAN and trunk reminders

- Access port: normally places an endpoint into one data VLAN.
- Trunk: carries multiple VLANs, usually using 802.1Q tags.
- Allowed VLAN list: VLANs permitted across a trunk; VLAN existence and spanning-tree state still matter.
- Native VLAN: untagged frames on an 802.1Q trunk are associated with the configured native VLAN on platforms using this model.
- Native VLAN mismatches can cause traffic to land in the wrong VLAN or trigger warnings.
- Inter-VLAN communication requires Layer 3 routing and suitable policy.
- An SVI is a virtual Layer 3 interface associated with a VLAN; its operational state depends on platform and VLAN/port conditions.

1.9 Routing reminders

- A router chooses a route using the destination IP and its routing table.
- Longest-prefix match generally selects the most specific matching route.
- A default route is used when no more specific route matches.
- A route existing in the table does not prove the full path or return path works.
- Check interface status, next-hop reachability, route source, ACLs/firewalls, and return routing.

1.10 Wi-Fi essentials

- SSID is the network name; it is not itself proof of authentication or internet access.
- Signal strength, interference, channel utilisation, distance, walls, client capability, and AP placement all affect performance.
- Strong signal bars do not guarantee gateway, DNS, or internet reachability.
- Authentication/encryption settings must match client and access point requirements.
- Compare wired and wireless results to isolate a radio-specific problem.

2. Troubleshooting playbooks

2.1 One device has no connectivity

- ☐ Record exact error and determine whether other devices are affected.
- ☐ Check adapter enabled state, cable/link lights, Wi-Fi association, and airplane mode.
- ☐ Inspect IP address, mask/prefix, gateway, and DNS.
- ☐ Look for an unexpected subnet or 169.254.x.x IPv4 address on Windows.
- ☐ Test the gateway, then a known remote IP, then a domain name.
- ☐ Compare with a working device on the same network.
- ☐ Make one evidence-based change and repeat the original test.

2.2 Wi-Fi connected but internet unavailable

- ☐ Check whether one or all devices on the SSID are affected.
- ☐ Confirm correct SSID and whether a captive portal is required.
- ☐ Inspect client addressing and test the gateway if permitted.
- ☐ Test a remote IP and DNS name separately.
- ☐ Check VPN/proxy settings and the router/AP uplink.
- ☐ If multiple clients fail, investigate shared equipment, DNS, or ISP service.

2.3 DHCP or wrong IP configuration

☐ Inspect address, prefix/mask, gateway, DNS, DHCP status, and lease details.

☐ Confirm link and VLAN before blaming the DHCP server.

☐ Check scope exhaustion, exclusions, reservations, relay/IP helper, and server availability.

☐ Renew the lease only when appropriate and compare before/after settings.

☐ Avoid arbitrary static IPs; check the address plan and possible conflicts.

2.4 Website fails by name

☐ Check spelling and service availability.

☐ Use nslookup to inspect the result and answering DNS server.

☐ Compare with an approved resolver or known-good device.

☐ Check VPN DNS, local resolver settings, filtering, and records.

☐ Clear local DNS cache only when stale cache is a plausible cause.

☐ A raw-IP browser test may be invalid because HTTPS hosting often depends on the hostname.

2.5 Slow or intermittent network

☐ Record time, affected apps/devices, and whether wired devices are affected.

☐ Separate latency, packet loss, throughput, and DNS delay.

☐ Compare near the access point versus the usual location.

☐ Inspect Wi-Fi interference, channel congestion, link speed, interface errors, and cables.

☐ Check heavy transfers as a controlled comparison.

☐ Treat ping/traceroute as clues; ICMP may be filtered or deprioritised.

2.6 VLAN or trunk issue

- ☐ Confirm access-port VLAN and VLAN existence/active status.
- ☐ Check trunk state and allowed VLAN list along the entire path.
- ☐ Check native VLAN configuration when untagged traffic is involved.
- ☐ Verify host IP/mask and expected gateway.
- ☐ Confirm the SVI/router interface exists and is operational.
- ☐ Then investigate routing, ACLs/firewalls, and return path.

2.7 Local network works; remote networks fail

- ☐ Confirm the destination is off-subnet according to the host mask.
- ☐ Inspect the default route and gateway.
- ☐ Test gateway reachability and inspect router routes if authorised.
- ☐ Check ACL/firewall policy, NAT where relevant, upstream routing, and return path.
- ☐ Use traceroute as supporting evidence, not proof of the exact fault.

2.8 A service port appears unreachable

- ☐ Confirm the destination name/IP and expected service port.
- ☐ Check whether the server is running and listening on that port.
- ☐ Test network reachability and TCP port connectivity from an authorised client.
- ☐ Check host firewall, network firewall, ACLs, NAT/port forwarding, and routing.
- ☐ Confirm the application binds to the expected interface and protocol.
- ☐ A successful TCP connection does not prove the application is healthy or secure.

2.9 Symptom-to-next-check decision table

Escalate when: impact is broad or increasing, a security incident is suspected, the fault involves shared infrastructure, the next action needs unapproved privileges/changes, or rollback is uncertain. Include scope, start time, evidence, tests and changes already attempted.

2.9 Symptom-to-next-check decision table

Observation	Prioritise checking
No cable link / no Wi-Fi association	Cable, NIC, power, radio, authentication, AP, switch port.
Link up but no valid address	DHCP, VLAN, relay, scope, static configuration.
Valid address but gateway unreachable	Mask, local VLAN, ARP/neighbour resolution, port security, gateway interface, host firewall.
Gateway works but remote IP fails	Routing, ACL/firewall, NAT, upstream link, remote service.
Remote IP works but domain lookup fails	DNS server, resolver configuration, DNS record, VPN DNS/filtering.
Only one application fails	Service health, port/protocol, proxy, TLS/certificate, application or endpoint firewall.
Only one VLAN is affected	Access VLAN, trunk allowed list, SVI, routing, policy.
Works after moving to another port	Original port configuration, VLAN, security policy, or physical fault.

2.9 Symptom-to-next-check decision table (part 2)

Interpret results carefully A failed ping does not prove a host is down; ICMP can be blocked. A successful ping does not prove the application port works. Test the specific layer or service in question.

2.10 Worked troubleshooting examples and self-check

The outputs below are illustrative examples, not captured results. Real wording and available fields vary by operating system, device, and software version. Use only on systems you are authorised to troubleshoot.

Example A: Windows has a 169.254.x.x address

Illustrative result: IPv4 Address . . . . . : 169.254.23.8; Default Gateway . . . . . : (blank).

Interpretation: Windows commonly uses an IPv4 link-local address when it has not obtained a DHCP lease. Check link/Wi-Fi association, VLAN, DHCP scope, relay/IP helper, and DHCP server path. Do not conclude that the DHCP server itself is faulty until the link and VLAN path have been checked. A manually configured address is an exception to this clue.

Example B: IP connectivity works but name lookup times out

Illustrative tests: ping 192.168.10.1 returns replies; nslookup portal.example.test times out against the configured DNS server.

Interpretation: the gateway test suggests local IP connectivity is working, but ping can be filtered and does not prove every service works. The DNS timeout directs the next investigation toward the configured resolver, DNS server reachability, UDP/TCP 53 policy, VPN DNS settings, and resolver service. Do not replace the hostname with an IP in a browser as a definitive test: HTTPS certificates and virtual hosting commonly depend on the hostname.

Example C: One VLAN cannot reach another

Check in order: client address and prefix → access VLAN → VLAN existence and trunk allowed list → SVI/router interface status → route and ACL/firewall policy → return path. If same-VLAN communication works but cross-VLAN communication fails, Layer 3 routing or policy becomes a stronger hypothesis, but verify the full path before changing configuration.

Self-check questions

1. A client has 169.254.8.4/16 and no gateway. Name two checks to perform before changing the address manually.
2. A host can ping its gateway but cannot resolve a domain. Which service should you investigate next, and why is this not absolute proof of a DNS-only fault?
3. Two hosts in the same VLAN communicate, but a host in another VLAN cannot reach them. Name three checks in a sensible order.
4. Why is a failed ping not enough to conclude that a remote application is down?
5. What should be recorded before and after a configuration change?

Answer key

- Check physical/Wi-Fi link and the expected VLAN; inspect DHCP lease/status, scope capacity, relay/IP helper, and server path.
- Check the configured DNS resolver and name-resolution path. A ping can be filtered, the DNS server may be unreachable, and other policies or application issues may coexist.
- Check the source host address/prefix, switch access VLAN and trunk path, then SVI/router interface and routing/ACL policy; verify the return path.
- ICMP may be filtered or deprioritised, while the application uses a different protocol/port. Test the actual service using an authorised method.
- Record symptoms, scope, baseline output, intended change, approval/rollback plan, commands/actions, verification results, and remaining issues.

3. Practical tools and templates

3.1 Command reference: Windows

3.1 Command reference: Windows

Command	Purpose	Interpretation note

Command	Purpose	Interpretation note
ipconfig /all	View adapter addressing, DHCP, gateway, DNS, MAC address.	Focus on the active interface, not disconnected adapters.
ipconfig /release	Release DHCP lease.	Can interrupt connectivity; use only when appropriate.
ipconfig /renew	Request DHCP lease.	If it fails, record the error and inspect DHCP path.
ipconfig /flushdns	Clear local DNS cache.	Does not fix incorrect DNS server or bad authoritative records.
ping <IP-or-name>	Test ICMP reachability.	Failure may be filtering; success does not test application port.
tracert <host>	Show responding hops toward destination.	Hops may be hidden or not respond.
nslookup <name>	Query DNS.	Note which DNS server answered and the returned record.
arp -a	View IPv4 ARP cache.	Entries are local-link neighbour information.
route print	Show routing table.	Look for default route and more-specific routes.
Get-NetIPConfiguration	PowerShell IP summary.	Useful for adapter and gateway checks.
Test-NetConnection <host> -Port 443	Test TCP reachability to a port.	Not a full HTTP/TLS or application test.
netstat -ano	View connections/listeners and process IDs.	Use tasklist /fi "PID eq <pid>" to map PID where authorised.

3.2 Command reference: Linux

3.2 Command reference: Linux

Command	Purpose	Note
ip addr	Interface addresses and state.	Modern replacement for many ifconfig use cases.
ip link	Link/interface state.	Does not prove upstream connectivity.
ip route	Routing table.	Check default route and next hop.
ip neigh	Neighbour/ARP cache.	IPv4 and IPv6 neighbour entries.
ping <host>	ICMP reachability.	May need Ctrl+C to stop; permissions/options vary.
tracepath <host> or traceroute <host>	Path diagnostic.	Availability depends on installed packages and permissions.
dig <name>	DNS query details.	Often more informative than nslookup if installed.
resolvectl status	Resolver status on systemd-resolved systems.	Not present on all Linux distributions.
ss -tulpn	Listening sockets and connections.	Some process details may require elevated privileges.
curl -I https://example.com	Inspect HTTP response headers.	Tests HTTP(S) behaviour, not just ICMP.

3.3 Cisco IOS / IOS XE read-only checks

3.3 Cisco IOS / IOS XE read-only checks

Command	Useful for
show ip interface brief	Interface IP and status overview.
show interfaces status	Port state and basic port details on supported switches.
show interfaces <interface>	Counters, errors, speed/duplex, and operational details.
show vlan brief	VLAN list and access-port membership.
show interfaces trunk	Trunk status and VLAN forwarding details.
show mac address-table	Learned MAC addresses and associated ports.
show ip route	Routing table and route sources.
show ip arp	IPv4 ARP entries on supported devices.
show running-config	Current active configuration; protect output.
show cdp neighbors detail	Cisco neighbour discovery if CDP is enabled.
show lldp neighbors detail	LLDP neighbour discovery if enabled.
show spanning-tree	Spanning Tree state; exact options vary by platform.

3.3 Cisco IOS / IOS XE read-only checks (part 2)

Command safety Start with read-only commands. Configuration commands can disconnect users or lock out management. Confirm device, interface, change scope, backup, rollback plan, and authorisation before making changes.

3.4 IP addressing plan template

3.4 IP addressing plan template

Field	Value to record
Site / purpose	
VLAN ID / name	
Network CIDR	
Subnet mask	
Gateway / SVI	
DHCP range	
Excluded / reserved addresses	
DNS servers / search suffix	
Static infrastructure addresses	
Routing / ACL notes	
Owner / review date	

3.5 Network inventory template

3.5 Network inventory template

Field	Value to record
Hostname / role	

Field	Value to record
Manufacturer / model	
Location / owner	
Management IP / prefix	
Default gateway / management VLAN	
Software / firmware version	
Interfaces / connected neighbour	
Critical dependencies	
Configuration backup location / date	
Support / warranty / change record	

3.6 Configuration review checklists

Switch access port

☐ Correct physical port and endpoint confirmed.

☐ Port enabled and link state understood.

☐ Correct access VLAN assigned and VLAN exists.

☐ Port security / authentication / edge protections match policy.

☐ Speed/duplex and errors checked if link is unstable.

☐ No unintended trunk configuration.

☐ Configuration saved according to lab/change policy.

☐ Connectivity verified and change documented.

802.1Q trunk

- ☐ Both ends intended to operate as trunks.
- ☐ Trunk operational state confirmed.
- ☐ Required VLANs exist and are allowed end-to-end.
- ☐ Native VLAN handling matches at both ends where applicable.
- ☐ Encapsulation and platform-specific settings verified.
- ☐ Spanning Tree state checked.
- ☐ Management reachability and rollback plan confirmed.

Router / Layer 3 interface

- ☐ Correct interface, address, and prefix.
- ☐ Interface administrative and operational state checked.
- ☐ Connected route appears as expected.
- ☐ Required routes/default route exist.
- ☐ Next-hop reachability checked.
- ☐ ACL/firewall policy reviewed in both directions.
- ☐ Return path and NAT requirements considered.
- ☐ Connectivity verified from relevant source networks.

DHCP scope

- ☐ Correct subnet and mask.
- ☐ Pool range excludes network, broadcast, gateway, and reserved static addresses.
- ☐ No unintended overlap with another pool or site.
- ☐ Default gateway and DNS options are correct.
- ☐ Scope has capacity and lease settings are appropriate.
- ☐ DHCP relay/IP helper and server path verified where needed.
- ☐ A client lease was tested and documented.

DNS change

- ☐ Correct zone and record name/type.
- ☐ Target value verified with the service owner.
- ☐ TTL and propagation expectations understood.
- ☐ Existing record and rollback value recorded.
- ☐ Authoritative and recursive answers checked as appropriate.
- ☐ Application tested by hostname after change.

Firewall / ACL rule review

- ☐ Business purpose and owner identified.
- ☐ Source, destination, protocol, and ports are as narrow as practical.
- ☐ Direction, rule order, statefulness, and implicit deny considered.
- ☐ Logging and expiry/review date set where appropriate.
- ☐ No broad any-to-any access without justified approval.
- ☐ Test plan, rollback, and post-change verification recorded.

3.7 Change record template

3.7 Change record template

Field	Record
Change ID / owner / date	
Reason and expected outcome	
Affected devices, users, and services	
Pre-change state / backup	
Planned steps	
Risk and impact	

Field	Record
Validation tests	
Rollback trigger and steps	
Actual result / verification	
Follow-up / documentation update	

3.8 Subnet calculator logic

For IPv4, a prefix /n means n network bits and 32−n host bits. For a conventional subnet, total addresses = 2^(32−n); usable hosts are usually total−2, except special cases such as /31 point-to-point links and /32 host routes.

3.8 Subnet calculator logic

Input	Example
IPv4 address	192.168.10.77
Prefix	/26
Network address	192.168.10.64
Broadcast address	192.168.10.127
First / last host	192.168.10.65 / 192.168.10.126
Total / conventional usable	64 / 62

- For real network planning, validate input format and prefix range (IPv4 prefix 0–32).
- Check that a proposed gateway lies inside the subnet and is not the network or broadcast address for conventional subnets.
- Check overlap against existing routes, VPNs, cloud VPC/VNets, and remote sites.
- Do not trust a calculator output without understanding whether it treats /31 and /32 specially.

4. Concept and protocol reference

IP addressing and subnetting

Function: Gives interfaces logical addresses and defines which destinations are local versus routed.

Related systems: IPv4/IPv6, subnet masks/prefixes, gateways, routes, DHCP.

Remember: The mask/prefix affects the host’s local-versus-remote decision.

Ethernet switching

Function: Moves frames inside a Layer 2 network using MAC learning and VLAN context.

Related systems: MAC address table, access ports, trunks, STP, ARP.

Remember: A switch does not automatically route between VLANs.

Routing

Function: Chooses a path for packets between IP networks.

Related systems: Routing table, connected/static/dynamic routes, next hop, longest-prefix match.

Remember: Forward path and return path both matter.

DNS

Function: Maps names to records used by clients and services.

Related systems: Recursive resolver, authoritative server, A/AAAA/CNAME/MX/PTR/TXT, caching, TTL.

Remember: A DNS failure and a network failure are different problems.

DHCP

Function: Automates client address and network configuration.

Related systems: Scopes/pools, leases, reservations, relay/IP helper, gateway and DNS options.

Remember: A client can have link but no usable lease.

ARP and IPv6 NDP

Function: Discover local-link neighbours for packet delivery.

Related systems: ARP for IPv4; ICMPv6 Neighbor Discovery for IPv6.

Remember: For a remote IPv4 host, a client normally resolves the gateway’s MAC—not the remote host’s MAC.

TCP and UDP

Function: Transport application data between process endpoints.

Related systems: Ports, sockets, TCP handshake/retransmission, UDP datagrams.

Remember: Port reachability is not the same as application health.

VLANs and trunks

Function: Separate Layer 2 broadcast domains and carry VLAN traffic between switches.

Related systems: 802.1Q tags, access ports, native VLAN, SVIs, inter-VLAN routing.

Remember: VLANs are logical segmentation, not a complete security boundary by themselves.

NAT/PAT

Function: Translates addresses and often ports between address domains.

Related systems: Private/public IPv4, connection tracking, port forwarding.

Remember: NAT is not encryption and does not replace firewall policy.

Firewalls and ACLs

Function: Enforce traffic policy based on rules and connection context.

Related systems: Stateful inspection, source/destination, protocol, port, rule order, logs.

Remember: A rule should be least-privilege, documented, and tested.

Wi-Fi

Function: Provides network access over shared radio spectrum.

Related systems: SSID, authentication, encryption, channels, APs, roaming, interference.

Remember: Association to an AP does not prove DHCP, DNS, or internet access.

VPN

Function: Creates an encrypted or otherwise protected logical connection across another network, depending on VPN type.

Related systems: IPsec, TLS-based VPNs, routes, DNS, identity and MFA.

Remember: A connected VPN can still have incorrect routes or DNS.

Network monitoring

Function: Collects telemetry and logs to show health, trends, and faults.

Related systems: SNMP, syslog, flow records, metrics, alerts, time synchronisation.

Remember: Good monitoring needs baselines and actionable alerts, not just data volume.

Worked example: how a host reaches a website

Example network: client 192.168.10.77/26, default gateway 192.168.10.65, and a website hostname that resolves to a remote IP address. This is a conceptual example; the actual path depends on DNS, routing, firewalls, and the service design.

6. The client checks the destination against its own IPv4 prefix. If the destination is outside 192.168.10.64/26, it sends the packet to its default gateway.
7. If the gateway MAC address is not already cached, the client uses ARP to learn the local-link MAC address for 192.168.10.65. It does not ARP for the remote website server across the internet.
8. The client sends an Ethernet frame to the gateway MAC. The IP packet still has the remote server as its destination; the router examines the destination IP and selects a route.

9. DNS resolution is a separate dependency: the client or its resolver must obtain an appropriate record. A cached answer may mean no new DNS query is needed.
10. Routers and security devices forward or filter traffic according to routes and policy. Replies also need a valid return path. HTTPS additionally depends on successful TLS negotiation and the correct hostname/certificate.

Key distinction: Ethernet source/destination MAC addresses are local-link delivery information and may change at each routed hop; the IP source/destination generally remain end-to-end unless NAT or another translation changes them.

5. Trusted IT resource directory

5.1 Structured learning and certification

Cisco Networking Academy — Introductory and career-aligned networking courses and learning pathways. https://www.netacad.com/

Cisco Skills for All — Free, self-paced IT, networking, cybersecurity, and career content where available. https://skillsforall.com/

Cisco Certifications — Official certification tracks, exam topics, and candidate information. https://www.cisco.com/site/us/en/learn/training-certifications/certifications/index.html

CompTIA Network+ — Vendor-neutral networking certification objectives and study information. https://www.comptia.org/certifications/network

Juniper Open Learning — Training and certification learning resources for Juniper technologies. https://learningportal.juniper.net/

Fortinet Training Institute — Network security and Fortinet product training. https://training.fortinet.com/

Microsoft Learn — Official Microsoft technical learning paths, including Windows networking and Azure. https://learn.microsoft.com/

AWS Skill Builder — AWS cloud training and certification preparation. https://skillbuilder.aws/

Google Cloud Skills Boost — Google Cloud learning paths and hands-on labs; some content may be paid. https://www.cloudskillsboost.google/

Red Hat Training — Linux and enterprise infrastructure learning resources. https://www.redhat.com/en/services/training-and-certification

5.2 Vendor documentation and technical references

Cisco Documentation — Official configuration guides, command references, release notes, and platform support. https://www.cisco.com/c/en/us/support/index.html

Cisco Command References — Find exact command syntax and supported options for Cisco platforms. https://www.cisco.com/c/en/us/support/all-products.html

Microsoft Learn: Networking — Windows Server, Windows client, Azure networking, and troubleshooting documentation. https://learn.microsoft.com/en-us/

AWS Networking and Content Delivery — VPC, routing, connectivity, DNS, load balancing, and network security docs. https://docs.aws.amazon.com/

Azure Architecture Center — Reference architectures and design guidance for Azure networking and infrastructure. https://learn.microsoft.com/en-us/azure/architecture/

Google Cloud networking docs — VPC, routes, firewall policies, hybrid connectivity, and load balancing. https://cloud.google.com/networking/docs

Juniper TechLibrary — Official Juniper product and configuration documentation. https://www.juniper.net/documentation/

Aruba Networking documentation — Official wireless and switching product documentation. https://www.arubanetworks.com/techdocs/

Ubiquiti Help Center — Guides for UniFi networking and related products. https://help.ui.com/

Linux man-pages — Reference pages for Linux system calls and utilities. https://man7.org/linux/man-pages/

Microsoft Sysinternals — Windows troubleshooting and administration utilities. https://learn.microsoft.com/en-us/sysinternals/

5.3 Standards, protocols, and registries

RFC Editor — Published RFCs covering Internet protocols, standards, and informational documents. https://www.rfc-editor.org/ Specific examples: RFC 1918 private IPv4 address allocation (https://www.rfc-editor.org/rfc/rfc1918.html); RFC 3021 /31 prefixes on IPv4 point-to-point links (https://www.rfc-editor.org/rfc/rfc3021.html).

IETF — Internet standards development, working groups, and standards process. https://www.ietf.org/

IANA Protocol Registries — Authoritative protocol numbers, service names, port numbers, and parameters. https://www.iana.org/protocols

IEEE 802 — LAN/MAN standards family including Ethernet and Wi-Fi; full standards may require access. https://www.ieee802.org/

RIPE NCC Academy — Internet number resources, routing, IPv6, and operational learning. https://academy.ripe.net/

APNIC Academy — Internet infrastructure, routing, IPv6, and network operator training for the Asia-Pacific region. https://academy.apnic.net/

ICANN Learn — DNS, domain names, and internet governance learning resources. https://learn.icann.org/

5.4 Simulators, packet analysis, and diagnostic tools

Cisco Packet Tracer — Network simulation for building topologies and practising Cisco-style configurations. https://www.netacad.com/courses/getting-started-cisco-packet-tracer/

Wireshark — Packet capture and protocol analysis software with documentation and display-filter references. https://www.wireshark.org/

Wireshark User Guide — Official guide to capture, display filters, protocol dissection, and analysis. https://www.wireshark.org/docs/wsug_html_chunked/

GNS3 — Network emulation platform; images for commercial devices may require separate licensing. https://www.gns3.com/

EVE-NG — Multi-vendor network emulation platform; check edition and image licensing requirements. https://www.eve-ng.net/

Nmap — Network discovery and security auditing tool; use only with explicit authorisation. https://nmap.org/

iPerf3 — Measures TCP/UDP throughput between endpoints. https://iperf.fr/

Speedtest by Ookla — Measures internet connection performance; results depend on server and conditions. https://www.speedtest.net/

MTR — Combines route tracing and repeated measurements on supported systems. https://www.bitwizard.nl/mtr/

PingPlotter — Visualises latency and packet-loss observations; product features/pricing vary. https://www.pingplotter.com/

SubnetOnline — Browser-based subnetting and IP tools; verify results for unusual prefixes. https://www.subnetonline.com/

IPcalc — IP address and subnet calculation utility; implementations vary. https://jodies.de/ipcalc

Quick tool finder: choose by task

Quick tool finder: choose by task

If you need to…	Try this first
Build a network visually	Packet Tracer or NetSim
Inspect real packets	Wireshark
Practise binary/subnetting	Cisco Binary Game or SubnettingPractice.com
Check DNS	Google Admin Toolbox Dig or DNSViz
Test a port	PortQry or curl
Measure throughput	iPerf3
Explore a route/latency	MTR or PingPlotter
Draw a topology	diagrams.net
Practise terminal commands safely	Linuxlab or Shellschool
Explore cloud network paths	SimAWS, then the cloud vendor’s official diagnostics

Use one tool at a time. Start with the free/simple option, check its limits, and return to the cheat sheets when you need a definition or command.

5.4A Network simulators and virtual labs

• Cisco Packet Tracer — Build a visual network and practise switching, VLANs, addressing, routing, DHCP, DNS, and packet flow. Free; Cisco account/sign-in may be required. https://skillsforall.com/resources/lab-downloads

• Cisco Packet Tracer: Getting Started — Short official introduction to the interface and first network. Free self-paced course. https://www.netacad.com/courses/getting-started-cisco-packet-tracer/1000

• NetSim — Browser-based network simulator with command-style configuration and animated traffic. Useful for seeing how packets move between hosts and devices. https://netsim.borck.education/

• Filius — Desktop simulator for basic networks, services, and data exchange; useful for visualising client/server behaviour. https://www.filius.org/

• GNS3 — Emulate more realistic network topologies. Setup is more involved; device images may require separate licensing. https://www.gns3.com/

• GNS3 documentation — Official setup and lab guides for GNS3. https://docs.gns3.com/

• EVE-NG — Multi-vendor virtual lab environment; best once basic concepts and computer resources are comfortable. https://www.eve-ng.net/

• Cisco Modeling Labs — Cisco virtual network lab environment. Check current edition, access, and image licensing terms. https://developer.cisco.com/modeling-labs/

5.4B Packet capture and protocol inspection

• Wireshark — Free, open-source packet analyser. Inspect DNS, DHCP, ARP, TCP handshakes, ICMP, and other protocols. https://www.wireshark.org/

• Wireshark User Guide — Official guide for capturing traffic, applying display filters, and reading packet details. https://www.wireshark.org/docs/wsug_html_chunked/

• Wireshark Display Filter Reference — Find field names and filter syntax for specific protocols. https://www.wireshark.org/docs/dfref/

• tcpdump — Command-line packet capture tool for Linux and Unix-like systems. https://www.tcpdump.org/

• Termshark — Terminal interface for exploring packet captures using tshark. https://termshark.io/

• CloudShark — Browser-based packet capture analysis. Check plan limits and never upload sensitive traffic captures. https://www.cloudshark.org/

5.4C Subnetting, binary, and IP address tools

- Cisco Binary Game — Short game to build speed converting decimal and binary. Free Cisco Learning Network account/sign-in required. https://learningnetwork.cisco.com/s/binary-game
- SubnettingPractice.com — Subnetting questions and timed practice for IPv4 masks, ranges, and hosts. https://subnettingpractice.com/
- SubnetOnline — Browser-based subnet calculators and IP utilities; use to verify your own calculation. https://www.subnetonline.com/
- IPcalc — Quick IPv4 network, mask, range, and broadcast calculation. https://jodies.de/ipcalc
- Calculator.net IPv6 Subnet Calculator — Browser-based IPv6 prefix and range calculator. https://www.tunnelsup.com/subnet-calculator/
- Hurricane Electric IPv6 Certification — Optional structured IPv6 tasks and checks, useful after IPv4 fundamentals. https://ipv6.he.net/certification/

5.4D Connectivity, ports, performance, and troubleshooting

• Microsoft PortQry — Check TCP/UDP port status and help diagnose service reachability on Windows. https://learn.microsoft.com/en-us/troubleshoot/windows-server/networking/portqry-command-line-port-scanner-v2

• Microsoft Sysinternals — Windows diagnostic utilities; TCPView helps inspect active connections. https://learn.microsoft.com/en-us/sysinternals/

• Nmap — Host discovery and port/service scanning. Scan only systems and networks you own or are explicitly authorised to test. https://nmap.org/

• iPerf3 — Measure throughput between two endpoints you control; useful for separating LAN performance from internet speed. https://iperf.fr/

• MTR — Repeated latency and route observations. Some routers filter or deprioritise probes, so interpret results carefully. https://www.bitwizard.nl/mtr/

• PingPlotter — Visualise latency and packet-loss observations; features and pricing vary by plan. https://www.pingplotter.com/

• Speedtest — Quick internet throughput and latency test; results vary with server and network conditions. https://www.speedtest.net/

• Advanced IP Scanner — Windows LAN discovery tool. Use only on networks where discovery is authorised. https://www.advanced-ip-scanner.com/

• Angry IP Scanner — Cross-platform IP and port scanner for authorised networks. https://angryip.org/

• curl documentation — Test HTTP/HTTPS endpoints, inspect response headers, and diagnose application-layer connections from a terminal. https://curl.se/docs/

5.4E DNS, domains, and internet routing

• Google Admin Toolbox Dig — Browser-based DNS lookups for common record types. https://toolbox.googleapps.com/apps/dig/

• MXToolbox SuperTool — Quick DNS, MX, SMTP, and domain diagnostics; some checks have limits. https://mxtoolbox.com/SuperTool.aspx

• DNSViz — Visual analysis of DNS configuration and DNSSEC validation. https://dnsviz.net/

• DNSChecker — Compare public DNS answers across locations; useful when checking whether a DNS change is visible. https://dnschecker.org/

• RIPE Atlas — Explore internet measurements from a global probe network; more advanced. https://atlas.ripe.net/

• BGP.Tools — Explore autonomous systems, IP prefixes, and BGP announcements. https://bgp.tools/

• Hurricane Electric BGP Toolkit — Look up autonomous systems, prefixes, and routing information. https://bgp.he.net/

• IANA Service Name and Port Registry — Authoritative registry for service names and registered port numbers. https://www.iana.org/assignments/service-names-port-numbers/service-names-port-numbers.xhtml

5.4F Diagramming and documentation

• diagrams.net — Create logical/physical network diagrams, topology maps, and rack layouts. https://www.diagrams.net/

• Excalidraw — Quick sketching for network paths and troubleshooting flows. https://excalidraw.com/

• Mermaid Live Editor — Create flowcharts and sequence diagrams from text; useful for packet paths and troubleshooting decisions. https://mermaid.live/

• NetBox — IP address management and infrastructure documentation platform; more useful for a homelab or later-stage projects. https://netboxlabs.com/oss/netbox/

5.4G Safe command-line and service practice

- Linuxlab — Browser-based Linux command practice; check sign-in and access requirements. https://linuxlab.io/en/p/linux/practice
- BashQuest — Simulated command-line tasks to build basic shell familiarity. https://www.bashquest.com/
- Shellschool — Simulated Linux and Windows command-line practice without changing your local system. https://shellschool.dev/en
- Docker Playground — Browser-based Docker environment when available; access requirements may change. https://labs.play-with-docker.com/
- Postman — Send HTTP requests and inspect responses; helpful for understanding application-layer behaviour. https://www.postman.com/
- CyberChef — Browser-based tool for encoding, decoding, and transforming text or byte data. https://gchq.github.io/CyberChef/
- OpenSSL — Command-line tools for inspecting certificates and working with TLS; use once comfortable with the terminal. https://www.openssl.org/

5.5 Useful IT communities and knowledge bases

Server Fault — Professional Q&A on system and network administration. https://serverfault.com/

Super User — Q&A for computer users, operating systems, and configuration issues. https://superuser.com/

Stack Overflow — Programming and scripting questions, including automation around infrastructure. https://stackoverflow.com/

Reddit r/networking — Networking industry discussions and operational perspectives; not authoritative documentation. https://www.reddit.com/r/networking/

Reddit r/ccna — CCNA study discussions and peer advice; verify technical claims. https://www.reddit.com/r/ccna/

Reddit r/homelab — Home lab ideas and infrastructure experience; verify product/version specifics. https://www.reddit.com/r/homelab/

Spiceworks Community — IT operations, administration, and troubleshooting discussions. https://community.spiceworks.com/

Packet Pushers — Networking podcasts, articles, and industry discussion. https://packetpushers.net/

The Networking Nerd — Networking explanations and commentary from an industry practitioner. https://networkingnerd.net/

Practical Networking — Explanations of routing, subnetting, and networking concepts. https://www.practicalnetworking.net/

5.6 Cloud, Linux, scripting, and automation

Learn Bash — Bash shell reference and learning materials. https://www.gnu.org/software/bash/manual/

PowerShell documentation — Official command and scripting documentation. https://learn.microsoft.com/en-us/powershell/

Python documentation — Language tutorial, standard library, and reference. https://docs.python.org/3/

Ansible documentation — Infrastructure automation, inventory, playbooks, and network automation modules. https://docs.ansible.com/

Terraform documentation — Infrastructure as code for cloud and other supported platforms. https://developer.hashicorp.com/terraform/docs

Git documentation — Version control for scripts, configs, notes, and infrastructure code. https://git-scm.com/doc

Docker documentation — Containers, networking, ports, and service connectivity. https://docs.docker.com/

Kubernetes networking concepts — Cluster networking, Services, DNS, network policies, and ingress concepts. https://kubernetes.io/docs/concepts/cluster-administration/networking/

Cloudflare Learning Center — Accessible explanations of DNS, CDN, TLS, DDoS, and internet infrastructure. https://www.cloudflare.com/learning/

AWS Well-Architected — Cloud architecture guidance including reliability, security, and operations. https://docs.aws.amazon.com/wellarchitected/latest/framework/welcome.html

5.7 Career, job skills, and keeping current

Cisco Learning Network — Certification community, exam preparation discussion, and study resources. https://learningnetwork.cisco.com/

Microsoft Technical Community — Product announcements, technical discussions, and community blogs. https://techcommunity.microsoft.com/

AWS Networking Blog — Cloud networking announcements, architecture, and implementation examples. https://aws.amazon.com/blogs/networking-and-content-delivery/

Cloudflare Blog — Internet infrastructure, security, performance, and protocol developments. https://blog.cloudflare.com/

SANS Internet Storm Center — Security observations and threat activity useful for security awareness. https://isc.sans.edu/

CISA Secure by Design / alerts — Security guidance and advisories; prioritise applicable official notices. https://www.cisa.gov/news-events/cybersecurity-advisories

Australian Cyber Security Centre — Australian cyber security guidance, alerts, and essential practices. https://www.cyber.gov.au/

Jobs and Skills Australia — Australian occupation and labour-market information. https://www.jobsandskills.gov.au/

SEEK Australia — Job listings to compare entry-level help desk, NOC, network support, and cloud roles. https://www.seek.com.au/

LinkedIn Learning — IT courses and career learning; access may require payment or library membership. https://www.linkedin.com/learning/

5.7 Career, job skills, and keeping current

Resource directory upkeep Links, course availability, certification objectives, and free access can change. Prefer current official pages. Community answers are useful clues, not a substitute for validating configurations against the exact device and software version.

Supplement: IPv6, wireless, platform commands and media

IPv6 address and configuration reference

IPv6 address and configuration reference

Prefix / mechanism	Meaning	Troubleshooting implication
2000::/3	Global unicast address space	Validate prefix, default route, upstream route and policy.
fe80::/10	Link-local	Required for many on-link functions; scope/interface is significant.
fc00::/7	Unique local; locally assigned prefixes commonly use fd00::/8	Private-use routing; not intended for public Internet routing.
ff00::/8	Multicast	IPv6 uses multicast for local control functions instead of IPv4-style broadcast.
::/128 and ::1/128	Unspecified and loopback	Special-use addresses, not ordinary LAN host assignments.

NDP versus ARP: ARP resolves IPv4 next-hop addresses on a local link. IPv6 NDP is implemented using ICMPv6 and includes Neighbor Solicitation/Advertisement plus Router Solicitation/Advertisement and Redirect functions. Filtering ICMPv6 broadly can break core IPv6 behaviour.

SLAAC versus DHCPv6: Router Advertisements communicate prefixes, flags and default-router information. SLAAC lets hosts form addresses from advertised prefixes. DHCPv6 can supply stateful addresses and/or additional configuration, depending on flags and deployment. The default router is normally learned from RA, not DHCPv6.

IPv6 address and configuration reference (part 2)

Platform	Useful read-only commands	What to inspect
Windows	`ipconfig /all`; `ping -6 <IPv6-address>`; `tracert -6 <IPv6-address>`; `netsh interface ipv6 show interfaces`; `Get-NetIPAddress -AddressFamily IPv6`	Address/prefix, interface state, default route and neighbour reachability.
Linux	`ip -6 addr`; `ip -6 route`; `ip -6 neigh`; `ping -6 <IPv6-address>` (syntax can vary)	Link-local and global addresses, routes, neighbour state and RA-derived configuration.
macOS	`ifconfig`; `networksetup -listallhardwareports`; `netstat -rn`; `scutil --dns`; `ipconfig getifaddr en0` for IPv4 where en0 is the target interface	Interface mapping matters; use interface-specific IPv6 route/diagnostic options as needed.

Wi-Fi poor performance or intermittent connectivity playbook

1. Scope the fault: one client, one location/AP, one SSID, or multiple users? Record times and recent changes.
2. Compare signal/RSSI, SNR if available, band, channel utilisation, interference, client capability, roaming and AP placement.
3. Test near the AP and on another client; compare wired performance to isolate the wireless segment.
4. Correlate client driver/power events with AP/controller logs and DHCP/DNS/gateway/application tests.
5. Change one variable at a time. Channel, transmit power, roaming, firmware and security changes require an approved plan and rollback.

Security modes: WPA2-Personal uses a shared passphrase; WPA2-Enterprise uses 802.1X/EAP and commonly RADIUS for central authentication. Enterprise faults may be certificate trust/expiry, time mismatch, EAP method, identity/policy, RADIUS reachability or server-side rejection. Prefer supported modern security settings and organisational policy.

Subnetting at larger prefixes

Subnetting at larger prefixes

Prefix	Mask	Total addresses	Traditional usable hosts*
/16	255.255.0.0	65,536	65,534
/17	255.255.128.0	32,768	32,766
/18	255.255.192.0	16,384	16,382
/19	255.255.224.0	8,192	8,190
/20	255.255.240.0	4,096	4,094
/21	255.255.248.0	2,048	2,046
/22	255.255.252.0	1,024	1,022
/23	255.255.254.0	512	510

*Traditional broadcast subnets subtract network and broadcast addresses. /31 is used for point-to-point IPv4 links under RFC 3021; /32 is a single-address route. The block-size method works for larger prefixes too: use 256 minus the relevant mask octet to find the subnet increment.

Service ports and physical media

SMTP: TCP 25 is commonly used for server-to-server relay; TCP 587 is message submission commonly secured with STARTTLS; TCP 465 is message submission using implicit TLS. Verify the actual service and provider configuration.

Service ports and physical media

Medium	Common capability reference	Distance / caveat
Cat5e	1 Gb/s	Up to 100 m compliant channel.

Medium	Common capability reference	Distance / caveat
Cat6	1 Gb/s to 100 m; 10 Gb/s on shorter channels	10GBASE-T often cited up to 55 m under qualifying conditions; verify installation and alien crosstalk.
Cat6A	10 Gb/s	Up to 100 m compliant channel.
OM3 multimode	10GBASE-SR	Typically up to 300 m for compliant 10GBASE-SR optics.
OM4 multimode	10GBASE-SR	Typically up to 400 m; optics and fibre specifications govern.

Escalation criteria and handover quality

Escalate when impact is broad/increasing, a security incident is suspected, the likely fault is in a shared/upstream service, an approved change window or privileged access is required, a fault persists after the agreed diagnostic boundary, or rollback is uncertain. Handover should include impact/scope, start time, recent changes, topology/VLAN/subnet where relevant, commands and outputs, tests run, findings, exclusions and risk.

Selected technical references

- IANA IPv6 Address Space: https://www.iana.org/assignments/ipv6-address-space/ipv6-address-space.xhtml
- RFC 4861, IPv6 Neighbor Discovery: https://www.rfc-editor.org/rfc/rfc4861
- RFC 4862, IPv6 Stateless Address Autoconfiguration: https://www.rfc-editor.org/rfc/rfc4862
- RFC 9915, DHCP for IPv6 (obsoletes RFC 8415): https://www.rfc-editor.org/rfc/rfc9915.html
- IANA Service Name and Port Number Registry: https://www.iana.org/assignments/service-names-port-numbers/service-names-port-numbers.xhtml
- RFC 6409, Message Submission: https://www.rfc-editor.org/rfc/rfc6409
- RFC 8314, Cleartext Considered Obsolete: https://www.rfc-editor.org/rfc/rfc8314
- Microsoft wireless troubleshooting: https://learn.microsoft.com/en-us/troubleshoot/windows-client/networking/wireless-network-connectivity-issues-troubleshooting

- Microsoft WPA2-Enterprise / 802.1X guidance: https://learn.microsoft.com/en-us/windows-server/networking/core-network-guide/cncg/wireless/a-deploy-8021X-wireless-access
- Cisco cabling/transceiver distance reference: https://www.cisco.com/c/en/us/products/collateral/interfaces-modules/transceiver-modules/data_sheet_c78-455693.html

6. Reference maintenance and study workflow

Quality rules for this reference

• Prefer official vendor documentation and published standards for commands, defaults, and protocol behaviour.

• Label vendor-specific behaviour rather than presenting it as universal networking law.

• Keep one main idea per section and put exceptions next to the rule they qualify.

• Avoid copying secrets into inventories: passwords, private keys, tokens, and sensitive configs should not be stored in ordinary notes.

• Use authorised systems only for scanning, packet capture, and connectivity testing.

• When a command changes state, mark it clearly and explain the risk before use.

7. Intermediate edition: diagnostic workbench

Use layered tests to isolate the failure domain. Compare a failing path with a known-good path, preserve command output, and change one variable at a time. Commands and configuration examples are platform-dependent; use an authorised lab or approved change window.

7.1 Structured fault isolation matrix

7.1 Structured fault isolation matrix

Test / observation	What it can establish	What it cannot establish	Next discriminating check
Interface shows up/up	Local interface and protocol state are up.	Correct VLAN, error-free media, end-to-end reachability.	Check counters, speed/duplex, VLAN, MAC learning and neighbour state.
Host can ping gateway	Some local L2/L3 path to the gateway responds.	DNS, upstream routing, firewall permission or application health.	Test a permitted remote IP, then DNS and target TCP port.
Remote IP works, name fails	IP path may be functioning while name resolution is not.	That every DNS record or application path is healthy.	Query the expected resolver and record type; check suffix, split DNS and cache.
TCP port test succeeds	A TCP handshake to that address and port completed at test time.	Authentication, TLS trust, application logic or sustained health.	Check service logs, TLS/certificate, authentication and application response.
Packet capture shows retransmissions	Observed packets are being retransmitted at the capture point.	The exact cause or where loss occurred without more evidence.	Compare capture points, interface errors, latency, path and server behaviour.

7.2 Subnetting worked example: /27

For `192.0.2.77/27`, the mask is `255.255.255.224`. The block size in the final octet is 256 − 224 = 32. The subnet boundaries are 0, 32, 64, 96 and so on. The address 77 falls in the 64–95 block.

7.2 Subnetting worked example: /27

Network	`192.0.2.64`
First / last conventional host	`192.0.2.65` – `192.0.2.94`
Broadcast	`192.0.2.95`
Usable hosts	30 (32 total addresses minus network and broadcast)

Sanity-check that the address is inside the range, the gateway is a usable address in the same subnet, and no overlapping subnet is already assigned. The conventional host calculation does not apply identically to every special prefix, such as IPv4 /31 point-to-point links or /32 host routes.

7.3 VLAN, trunk and SVI troubleshooting sequence

1. On the endpoint, verify the expected subnet, gateway and physical switch port.
2. On the access switch, check the interface state, access VLAN, errors and MAC address learning.
3. On the trunk path, verify trunk operational state, allowed VLAN list, native VLAN agreement and that the required VLAN exists and is active at each hop.
4. On the Layer 3 switch, verify the SVI exists, has the intended IP address, is not administratively down, and is operational. An SVI may remain down when VLAN or port conditions are not satisfied.
5. Check routing and ACL/policy only after confirming the local VLAN and gateway path.

```
show interfaces status
show vlan brief
show interfaces trunk
show mac address-table dynamic
```

```
show ip interface brief
show ip route
```

These are read-only Cisco IOS/IOS XE examples; available syntax varies by platform and release. Avoid applying a configuration command simply because a show command looks unusual. Capture the current state and compare with the intended design.

7.4 DHCP and DNS diagnostic checkpoints

DHCP: DHCPv4 commonly follows Discover, Offer, Request, Acknowledgement (DORA). If a client lacks the expected lease, check the client VLAN, relay/helper path where used, server scope availability, exclusions, relay reachability and server logs. Do not assume every failure is a full DHCP server outage.

DNS: Compare the failing query against the configured resolver and a known-good client on the same network. Check A and AAAA records as appropriate, search suffixes, split-horizon DNS, TTL/cache and whether the resolver can reach authoritative services. Avoid changing global DNS settings before confirming the intended resolver policy.

```
ipconfig /all
nslookup -type=A app.example.test
nslookup -type=AAAA app.example.test
Test-NetConnection app.example.test -Port 443
```

The hostname and domain suffix above are examples. Replace them with approved test names; do not use fictitious example domains as real service targets.

7.5 Packet capture: filter, compare, infer cautiously

7.5 Packet capture: filter, compare, infer cautiously

Question	Useful Wireshark display filter	Interpretation caution
Are ARP requests unanswered?	`arp`	Capture location and VLAN visibility matter; some traffic may be offloaded or not visible.

Question	Useful Wireshark display filter	Interpretation caution
Is DNS being queried and answered?	`dns`	Encrypted DNS or a different resolver path may not appear as ordinary DNS.
Is TCP setup failing?	`tcp.flags.syn == 1`	A SYN without a visible reply may reflect filtering, path asymmetry or capture loss.
Are ICMP errors present?	`icmp || icmpv6`	Absence of ICMP is not proof of a healthy or failed path.

Capture only traffic you are authorised to inspect. Minimise captured data, protect files, and avoid collecting credentials or unrelated personal information.

7.6 Scenario: one VLAN cannot reach a shared application

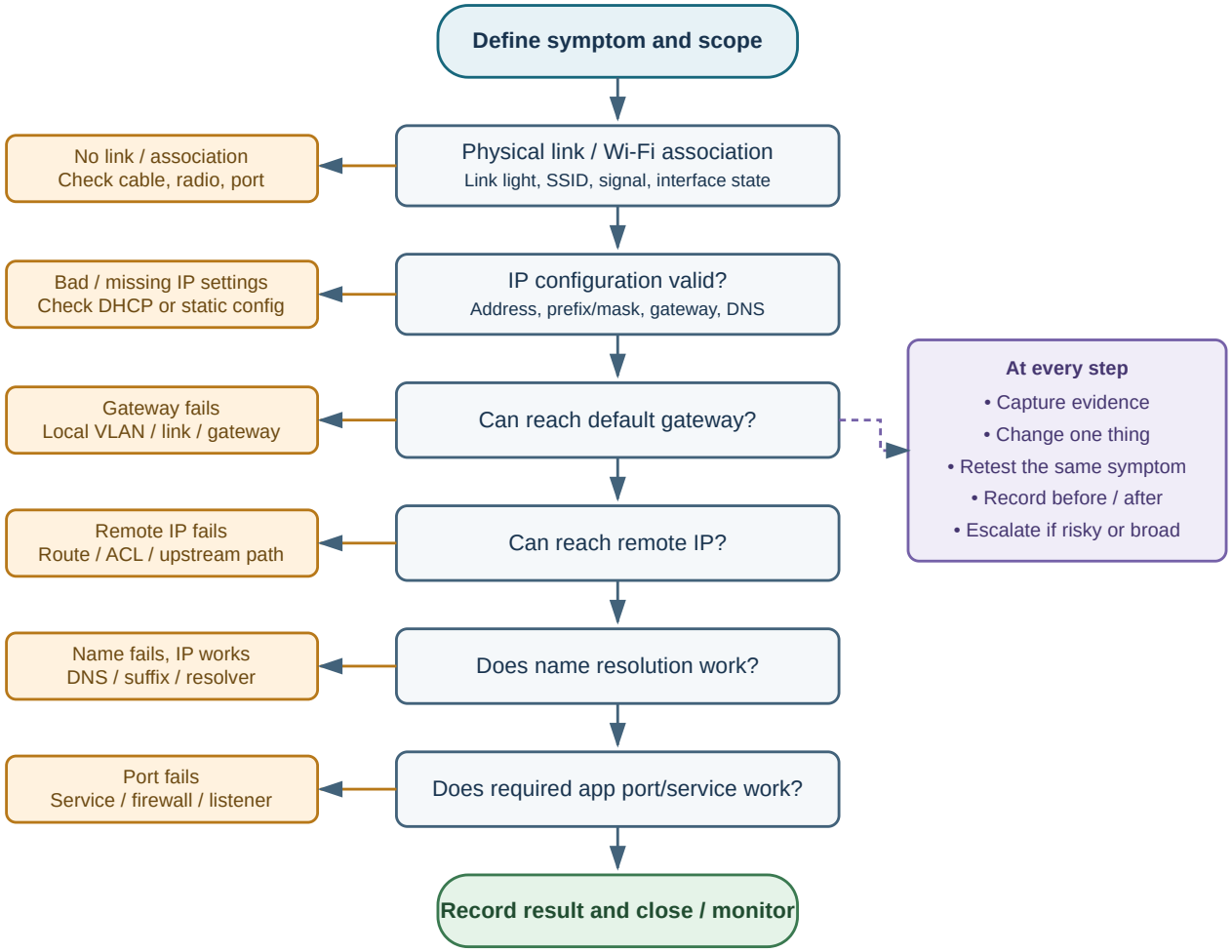
Users in VLAN 30 can reach their SVI but cannot reach a server in VLAN 50; users in VLAN 50 can reach the server locally. Compare the failing and working path. Check the VLAN 30 gateway SVI, routing table, return route from the server network, ACLs/firewall policy in both directions, DNS answer, and target TCP port. A successful ping from VLAN 30 is not required for TCP to work if ICMP is filtered, so use multiple evidence sources. Record the exact source, destination, protocol, port and timestamp.

7.7 Escalation package

Include scope and business impact, topology/path, source and destination, exact timestamps and timezone, recent changes, command outputs, packet capture only when approved, known-good comparison, steps already attempted, and a safe rollback suggestion. Escalate before disruptive changes or when evidence points to a shared upstream service.

Network troubleshooting flow

Follow the path from the simplest local checks toward routing, name resolution and application service. Use the failed branch to choose the next investigation, not to make an unsupported conclusion.



Use this as a sequence, not a guarantee: ICMP may be blocked, a host may use IPv6, and application failures can occur even when lower layers work. Compare known-good devices and paths. Stop before disruptive changes unless authorised.

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Network troubleshooting flow

This document was developed with assistance from AI tools for drafting, organisation and refinement, and reviewed by the author against their own networking knowledge and learning. Verify platform-specific syntax and current vendor guidance before making changes. Use only on systems you are authorised to inspect or administer.

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